Calibrated Single-Pass Action Selection for Mobile GUI Agents

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Abstract

Mobile GUI agents are usually built on vision-language models that decode a reasoning trace and an action string at every step. Such agents must parse their own output, pay a sequential decoding cost per step, and expose no calibrated probability for the action they take. We study the alternative of a *decision model*, which reads the screen state once and returns a probability distribution over typed answers in a single forward pass without generating tokens. We fine-tune the 2B decision model d1-2B on AndroidControl and evaluate the resulting agent, DMA-2B, on AndroidControl and on the official AndroidWorld protocol. DMA-2B reaches a step success rate of 89.3% on AndroidControl-Low and 61.7% on AndroidControl-High (from 59.8% and 40.2% before fine-tuning), and a task success rate of 21.7% on AndroidWorld, at a median latency of 135 ms per decision. Its action probability ranks its own errors (AUROC 0.758 on 8,444 AndroidControl-High steps), and a split-conformal threshold on that probability keeps the error rate of the accepted steps below 20% while acting on 49% of them. On AndroidWorld, the probability attached to the model's own termination decision separates correct from premature terminations (AUROC 0.746): escalating terminations below 0.9 would intercept 36 of 49 premature ones at the cost of 11 of 26 correct ones. We further show that per-step accuracy overstates task success by roughly a factor of three, and estimate that two fifths of the AndroidWorld failures require computation that a single forward pass does not provide.

1. Introduction

Most mobile GUI agents are generative. At each step a vision-language model (VLM) reads a screenshot, writes a thought, and emits an action as text, which the harness parses and executes (Rawles et al., 2025; Qin et al., 2025; Ye et al., 2025; Xu et al., 2026). This design has three costs that matter for an agent running on a device. First, the action is a string, so a malformed or out-of-schema output is a runtime error. Second, decoding is sequential, and its cost grows with the length of the reasoning trace. Third, the agent does not expose a calibrated probability for the action it selected, so the surrounding software cannot decide whether to act, to ask the user, or to stop.

A decision model removes all three costs by construction (TypeSafe AI, 2026a; Liquid AI, 2026). It receives a state and a set of typed questions, and returns, for each question, a probability distribution over a closed answer set in one forward pass. Its output cannot violate the schema, its cost is one forward pass, and each answer comes with a probability. Whether such a model can operate a phone, and whether its probabilities are informative enough to decide when it should not act, has not been measured.

This paper reports that measurement. We define a decision interface for mobile control, fine-tune the 2B decision model d1-2B (Liquid AI, 2026) on AndroidControl (Li et al., 2024), and evaluate the resulting agent, DMA-2B, on AndroidControl and on AndroidWorld (Rawles et al., 2025). Our contributions are the following.

1. A single-pass decision interface for mobile GUI control (Section 3), in which the action, the target element, the scroll direction, the app and any typed text are answered jointly from one forward pass, and the text state is produced by the same renderer at training and at inference.

2. An evaluation on AndroidControl in the standard Type / Grounding / SR format against published models, including the untrained base model, and on the official AndroidWorld protocol (Section 5).

3. An analysis of the model's confidence as a control signal (Section 6): how well it ranks errors before and after fine-tuning, what selective risk a split-conformal threshold achieves, and how well it separates correct from premature task termination on AndroidWorld.

4. A decomposition of the gap between per-step accuracy

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and task success, based on a trace-level reading of all failed AndroidWorld tasks (Section 7).

2. Related Work

Mobile GUI agents. AndroidControl (Li et al., 2024) and AndroidWorld (Rawles et al., 2025) are the two standard Android benchmarks: the first scores individual steps against human demonstrations, the second scores task completion in a live emulator. Recent agents are VLMs trained with supervised fine-tuning on GUI trajectories (Wu et al., 2025; Xu et al., 2025; Qin et al., 2025) and, increasingly, with reinforcement learning from rule-based or environment rewards (Lu et al., 2026; Luo et al., 2025; Liu et al., 2025; Xiao et al., 2025; Xu et al., 2026). Several of them are small: UI-TARS-2B, OS-Atlas-4B, UI-R1-3B, GUI-R1-3B and InfiGUI-R1-3B all report AndroidControl results (Table 1). All of these models generate text, and several generate an explicit reasoning trace before the action. Multi-agent systems add planners, reflectors and verifiers around the VLM (Ye et al., 2025; Dai et al., 2025; Lai et al., 2025).

Decision models. TypeSafe's Jev and LiquidAI's d1 family answer typed questions (choice, yes/no, score, span, box) in one pass and return probabilities rather than text (TypeSafe AI, 2026a; Liquid AI, 2026). TypeSafe's documentation advises keeping "math in code, which includes counting and date comparison" (TypeSafe AI, 2026b); our failure analysis arrives at the same boundary from the agent side.

Calibration and selective prediction. A classifier is calibrated when its confidence matches its accuracy (Guo et al., 2017). Selective prediction lets a model abstain where its confidence is low and trades coverage for risk (Geifman & El-Yaniv, 2017). Learn-then-Test (Angelopoulos et al., 2021) and conformal risk control (Angelopoulos et al., 2024) turn a confidence score into a threshold with a finite-sample guarantee on the risk of the accepted decisions. CORA (Feng et al., 2026) applies conformal risk control to mobile GUI automation, gating the actions of a generative agent with a separate guardian model. We apply the same statistical tools to the probability that the acting model itself produces.

3. Method

3.1. Problem formulation

At step t the agent observes a screenshot x_t and the accessibility tree of the screen, and knows the goal g and its own previous actions $h_{<t}$. A deterministic renderer turns $(g, h_{<t}, \text{tree})$ into a text state z_t that lists the goal, the current app, the previous actions with their observed effects, and the visible elements $\mathcal{E}_t = \{e_1, \ldots, e_{n_t}\}$ with label, type, state and position. The model receives $s_t = (x_t, z_t)$ and a fixed set of typed questions, and returns in one forward pass

$$p(a \mid s_t),\ p(e \mid s_t),\ p(d \mid s_t),\ p(m \mid s_t),\ p(\sigma \mid s_t), \quad (1)$$

where a ranges over ten action types (click, long press, type, scroll, wait, back, home, open app, done, answer), $e \in \mathcal{E}_t$ is the target element, d the scroll direction, m the app to open, chosen from the installed apps, and σ a span of z_t that is copied as the text to type or the answer to report. The executed action is assembled from the arg-max answers: $\hat{a} = \arg\max_a p(a \mid s_t)$ together with the answers that $\hat{a}$ requires. Because spans are copied from the text state, the model cannot produce text that appears only in the pixels (Section 7).

3.2. Model and training

The decision model is d1-2B (Liquid AI, 2026), a 1.92B-parameter vision-language decision model whose readout heads project the backbone state onto the token groups of each question's options. We fine-tune all parameters except the tied input embedding and output head, including the vision encoder, with a cross-entropy loss on every question that the gold action defines. We call the fine-tuned model DMA-2B.

The training set contains 37,977 decisions from the AndroidControl training split (Li et al., 2024), rebuilt from the original release with all test episodes excluded and restricted to task kinds that also occur in AndroidWorld. Each history line carries the effect observed after the action ("screen changed" or "no change"). We add rows in which an action had no effect and the gold answer is a different action, rows that end an episode with an answer, and, on a fixed half of the rows, the AndroidControl step instruction. We also add 1,736 rows that map goals produced by AndroidWorld's task generator to the app to open. These rows cover 97 of the 116 evaluated task templates; they contain no AndroidWorld screen, trajectory or success checker, but they do expose the model to AndroidWorld's goal phrasing. Training runs for one epoch (2,482 steps, 16 rows per step, learning rate 10^{-5}).

3.3. Confidence and escalation

We use the probability of the selected action, $c(s_t) = p(\hat{a} \mid s_t)$, as the step confidence, and compare it with two combinations that include the element probability $p(\hat{e} \mid s_t)$ on steps whose action targets an element (their product and their minimum). An escalation gate acts if $c(s_t) \geq \tau$ and asks otherwise. For n scored steps with correctness

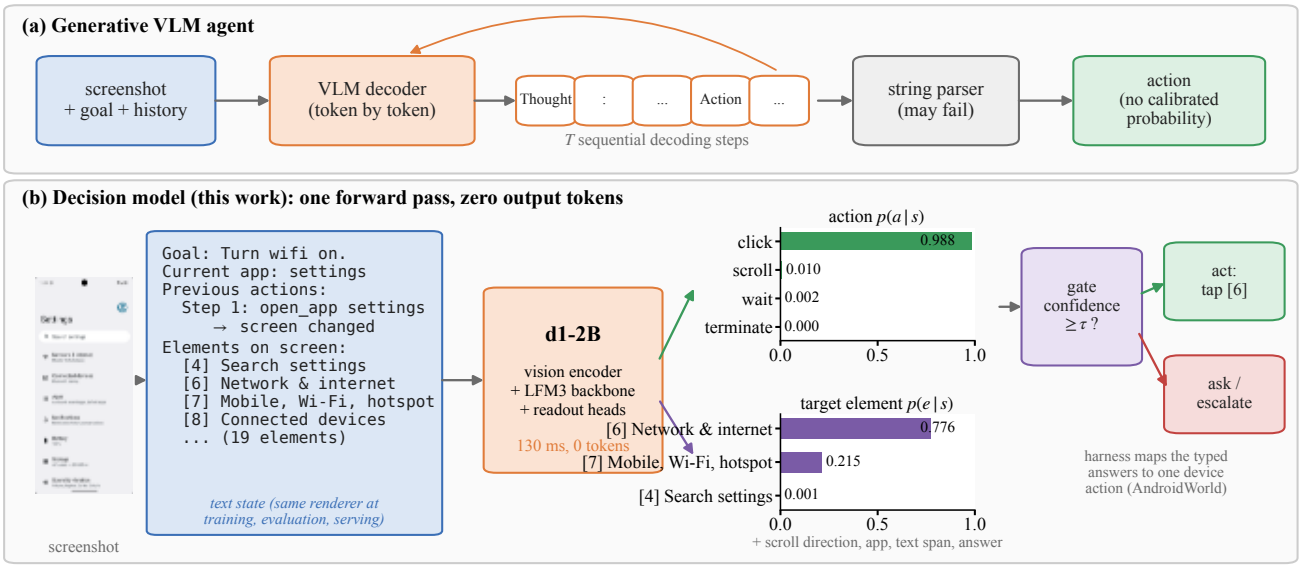


Figure 1. Generative agents and the decision interface. (a) A generative VLM agent decodes a reasoning trace and an action string, which is parsed into an action. (b) The decision model reads the screenshot and a text state once and returns a distribution for every typed question; a confidence gate decides between acting and escalating. The screenshot, the state and both distributions are taken from one recorded AndroidWorld step (task "Turn wifi on.", step 2); the model took 130 ms for this decision.

$y_i \in \{0, 1\}$, coverage and selective risk are

$$\mathrm{cov}(\tau) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[c_i \geq \tau], \tag{2}$$

$$R(\tau) = \frac{\sum_i \mathbf{1}[c_i \geq \tau]\,(1 - y_i)}{\sum_i \mathbf{1}[c_i \geq \tau]}. \tag{3}$$

To choose τ for a target risk δ we follow Learn-then-Test with fixed-sequence testing (Angelopoulos et al., 2021): on a calibration set we lower τ from the most confident decision and stop before the first threshold at which the one-sided Clopper–Pearson upper bound on $R(\tau)$ at level $\alpha = 0.05$ exceeds δ.

4. Experimental Setup

AndroidControl. The test split has 8,444 steps in 1,543 episodes (Li et al., 2024). In the Low setting the model sees the step instruction; in the High setting it sees only the goal. In both settings the history is the demonstrator's, so every step is evaluated from the correct prefix. We score with the evaluation code of InfiGUI-R1 (Liu et al., 2025) without modification and report the three usual metrics: *Type*, the action-type accuracy; *Grounding*, the share of click and long-press steps with the correct type whose predicted point falls inside the target element; and *SR*, the step success rate. We additionally report the *episode success rate* (ESR), the share of episodes in which every step is correct.

AndroidWorld. We run the 116 tasks with seed 30 and one attempt per task under the official success checkers (Rawles et al., 2025). The agent decides by itself when the task is done; there is no oracle stop. The *task success rate* (TSR) is the sum of task scores divided by the number of scored tasks, so a partial score of 0.5 counts as half a success. One task (OsmAndMarker) failed during environment setup and is excluded, leaving 115 scored tasks. Decoding is greedy, but the emulator is not deterministic: in an earlier check, 6 of 20 replays on the same host produced different trajectories.

Confidence metrics. We measure how well a confidence score ranks correct above incorrect steps by AUROC, its calibration by the expected calibration error (ECE, 15 equal-width bins) (Guo et al., 2017), and its use for selection by the risk at fixed coverage and the area under the risk–coverage curve (AURC) (Geifman & El-Yaniv, 2017).

5. Results

5.1. AndroidControl

Table 1 places DMA-2B among published models. Fine-tuning raises step success from 59.8% to 89.3% on AndroidControl-Low and from 40.2% to 61.7% on AndroidControl-High, and episode success from 18.8% to 60.4% (Low) and from 9.4% to 22.3% (High). On AndroidControl-Low, DMA-2B matches UI-TARS-2B (89.3% SR) and exceeds OS-Atlas-4B and Aguvis-7B as reported by InfiGUI-R1, while InfiGUI-R1-3B and the 7B–

Table 1. Results on the AndroidControl test set (Li et al., 2024) (8,444 steps). Type: action-type accuracy; Grounding: target accuracy on click and long-press steps with the correct type; SR: step success rate. Rows in the first three groups are copied from the cited tables, which use different prompts, action spaces and evaluation code; they are comparable with each other and with ours only approximately, and the same model can differ across groups (e.g. OS-Atlas-4B). Our rows use the InfiGUI-R1 scorer. Bold: best; underline: second best; "–": not reported.

		AndroidControl-Low			AndroidControl-High		
Model	Size	Type	Grounding	SR	Type	Grounding	SR
Reported by InfiGUI-R1 (Liu et al., 2025)							
Claude*	–	74.3	0.0	19.4	63.7	0.0	12.5
GPT-4o (OpenAI, 2024)	–	74.3	0.0	19.4	66.3	0.0	20.8
Aria-UI (Yang et al., 2024)	–	–	87.7	67.3	–	43.2	10.2
OS-Atlas-4B (Wu et al., 2025)	4B	91.9	83.8	80.6	84.7	73.8	67.5
Aguvis-7B (Xu et al., 2025)	7B	–	–	80.5	–	–	61.5
Aguvis-72B (Xu et al., 2025)	72B	–	–	84.4	–	–	66.4
UI-R1-3B (Lu et al., 2026)	3B	94.3	82.6	–	–	–	–
GUI-R1-3B (Luo et al., 2025)	3B	–	–	–	58.0	56.2	46.6
GUI-R1-7B (Luo et al., 2025)	7B	–	–	–	71.6	65.6	51.7
UI-TARS-2B (Qin et al., 2025)	2B	98.1	87.3	89.3	81.2	78.4	68.9
InfiGUI-R1-3B (Liu et al., 2025)	3B	96.0	93.2	92.1	82.7	74.4	71.1
Reported by GUI-R1, shared zero-shot prompt (Luo et al., 2025)							
OS-Atlas-4B (Wu et al., 2025)	4B	64.6	71.2	40.6	49.0	49.5	22.8
Qwen2.5-VL-3B (Bai et al., 2025)	3B	62.0	74.1	59.3	47.8	46.5	38.9
Qwen2.5-VL-7B (Bai et al., 2025)	7B	83.4	87.1	62.5	68.7	59.7	47.1
UI-R1-3B (Lu et al., 2026)	3B	79.2	82.4	66.4	57.9	55.7	45.4
GUI-R1-3B (Luo et al., 2025)	3B	83.7	81.6	64.4	58.0	56.2	46.6
GUI-R1-7B (Luo et al., 2025)	7B	85.2	84.0	66.5	71.6	65.6	51.7
Reported by MobileRL (Xu et al., 2026)							
UI-TARS-7B (Qin et al., 2025)	7B	98.0	89.3	90.8	83.7	80.5	72.5
UI-TARS-72B (Qin et al., 2025)	72B	98.1	90.8	91.3	85.2	81.5	74.7
UI-Genie-7B (Xiao et al., 2025)	7B	98.1	89.3	94.3	83.5	82.9	74.5
UI-Genie-72B (Xiao et al., 2025)	72B	**98.3**	**95.4**	**94.8**	84.9	**86.3**	**77.0**
MobileRL-7B (Xu et al., 2026)	7B	96.9	92.4	91.4	86.3	71.5	71.3
MobileRL-9B (Xu et al., 2026)	9B	97.9	94.6	94.3	**87.0**	74.5	73.9
This work: decision model, zero output tokens, InfiGUI-R1 scorer							
d1-2B (base) (Liquid AI, 2026)	2B	80.0	76.9	59.8	63.6	63.1	40.2
DMA-2B (ours)	2B	96.6	90.1	89.3	74.3	78.4	61.7

72B models of the MobileRL group remain ahead. On AndroidControl-High, DMA-2B is above every model evaluated by GUI-R1 under its shared zero-shot prompt, including the 7B models, and level with Aguvis-7B (61.5%); UI-TARS-2B (68.9%), OS-Atlas-4B as reported by InfiGUI-R1 (67.5%) and InfiGUI-R1-3B (71.1%) remain ahead. Its High grounding (78.4%) equals that of UI-TARS-2B and exceeds those of InfiGUI-R1-3B and of both MobileRL models, so the remaining High gap lies mainly in the action type. Every other model in Table 1 generates text; DMA-2B generates none.

5.2. AndroidWorld

DMA-2B solves 24 of the 115 scored tasks and receives half credit on two more, a TSR of 25/115 = 21.7%. Table 2 breaks the result down by the difficulty labels and optimal step counts of the AndroidWorld task metadata, and by complexity level, defined as in MobileRL as the ceiling of a task's complexity value. Success is concentrated in easy and level-1 tasks. On the tasks it solves, the agent uses more steps than the optimum (5.5 against 3.2 on easy tasks, 6.7 against 5.0 on medium tasks).

Table 2. AndroidWorld task success of DMA-2B (seed 30, pass@1, 115 scored tasks). Steps: mean steps taken; Opt.: mean optimal steps from the task metadata.

Split	Tasks	Score	TSR (%)	Steps	Opt.
Easy	61	21.0	34.4	8.5	5.0
Medium	36	3.0	8.3	13.0	9.0
Hard	18	1.0	5.6	19.3	18.5
Level 1	45	19.0	42.2	–	–
Level 2	40	5.5	13.8	–	–
Level 3	9	0.0	0.0	–	–
Level $\geq$4	21	0.5	2.4	–	–
All	115	25.0	21.7	11.6	8.4

Table 3 lists AndroidWorld results reported by other systems. They are not a ranking: none was obtained with our

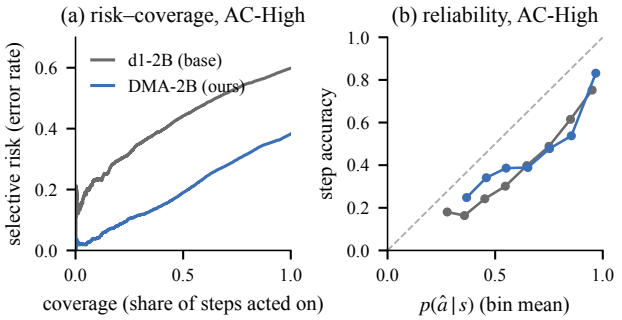


Figure 2. AndroidControl-High, base model and DMA-2B. (a) Error rate among the steps the gate accepts, as the threshold is lowered. (b) Step accuracy against binned confidence; the dashed line is perfect calibration.

harness, and the systems differ in size, number of model calls per step, generated reasoning, extra components and use of AndroidWorld-derived training data. The closest settings, single-model 2B agents without tools, are MAI-UI-2B (49.1%) (Zhou et al., 2025) and UI-Venus-1.5-2B (55.6%) (Venus Team et al., 2026); both generate text, and we have not run a generative model of the same size in our harness.

Latency. Over 1,335 AndroidWorld steps the model takes 135 ms per decision at the median and 181 ms at the 95th percentile (one forward pass on a shared 24 GB GPU, served over HTTP, no tokens generated). The emulator step takes 1,624 ms at the median, so the device, not the model, dominates the agent loop.

6. Confidence as a Control Signal

6.1. Does the confidence rank errors?

Table 4 and Figure 2 compare the step confidence of the base model and of DMA-2B on AndroidControl-High. Fine-tuning improves both accuracy and ranking: AUROC rises from 0.721 to 0.758 and the risk at 50% coverage falls from 44.2% to 19.0%. Both models are over-confident (mean confidence exceeds accuracy by 0.23 and 0.19), so the raw probability is a ranking signal rather than a calibrated estimate of correctness. For the action type alone, the probability of DMA-2B is close to calibrated (ECE 0.073) and ranks type errors with AUROC 0.800.

The best way to combine the action and element probabilities depends on how correctness is defined. Under the InfiGUI-R1 scorer, which accepts a click inside any of the candidate boxes of the target, the action probability alone ranks best (AUROC 0.758, against 0.701 for the product with the element probability and 0.717 for the minimum). Under a stricter scorer that requires the point to lie inside the smallest gold element, accuracy falls to 52.5% and the order reverses (0.715, 0.721, 0.732); on the 827-step validation split, scored the same strict way, the product reaches 0.757 against 0.701 for the action probability. A gate must therefore be calibrated against the success criterion of the deployment, not against a proxy.

6.2. Risk-controlled gating

We split the AndroidControl-High test episodes into two halves 200 times, choose τ on one half with the procedure of Section 3.3, and apply it to the other. For a target risk $\delta = 20\%$ a threshold exists in every split; the gate then acts on 49.1% of the held-out steps with a mean error rate of 18.6%. For $\delta = 15\%$ a threshold exists in 169 of 200 splits (coverage 33.3%, risk 12.4%), and for $\delta = 10\%$ in only 55 (coverage 14.5%, risk 7.2%). The guarantee concerns the expected risk; on a half-size test set the realised risk still exceeds δ in 11–15% of the splits, and splitting by step rather than by episode gives a similar rate. A threshold chosen on the validation split instead is valid but very conservative: for $\delta = 20\%$ it accepts only 10.0% of the test steps (risk 4.1%), because the validation split is scored by the strict criterion and has a different action mix.

6.3. Confidence at termination on AndroidWorld

The most frequent AndroidWorld failure is a premature "done". Of the 115 scored episodes, 75 end with the model's own termination; 26 of these earn credit and 49 do not. The probability that the model assigns to its termination separates the two groups (AUROC 0.746; mean 0.884 against 0.704; Figure 3). Escalating every termination with $p(\text{done} \mid s) < 0.9$ would intercept 36 of the 49 premature terminations and 11 of the 26 credited ones; at 0.95 the counts are 40 and 13. These thresholds are chosen on the same 75 episodes and are descriptive, and what an intercepted episode would achieve depends on the answer to the escalation, which we do not measure. Two further signals behave similarly: the lowest step confidence in an episode predicts task success with AUROC 0.777, and step confidence predicts whether the action changed the screen with AUROC 0.746 (1,119 steps).

7. Where the Gap Comes From

Per-step accuracy and episodes. The AndroidControl scorer feeds the demonstrator's history at every step and never checks the goal, so errors do not compound. Figure 4 shows the episode success rate by episode length: DMA-2B completes 59% of one-step episodes but 5% of episodes with ten or more steps. Its High episode success rate (22.3%) is close to its AndroidWorld TSR (21.7%), although the two differ in kind: AndroidWorld requires the agent to stop by itself and contains 13 apps without training goals, but accepts any path to the goal state. The Android-

Table 3. AndroidWorld task success rates reported by each system's authors, for context only. None was obtained with our harness. Calls: model calls per step; AW data: AndroidWorld-derived training data; "–": not checked in the source.

System	Size	Calls	Text	AW data	Extra components	TSR (%)
DMA-2B (ours)	2B	1	no	goal-to-app rows from the task generator	none	**21.7**
M3A, GPT-4 Turbo (Rawles et al., 2025)	–	≥2	yes	no	step summary as memory	30.6
UI-TARS (Qin et al., 2025)	–	1	yes	not stated	–	46.6
AndroidGen, GPT-4o (Lai et al., 2025)	–	>1	yes	trajectories collected in AndroidWorld	planner, retrieval, checks	46.8
MAI-UI-2B (Zhou et al., 2025)	2B	1	yes	not stated	none	49.1
UI-Venus-1.5-2B (Venus Team et al., 2026)	2B	1	yes	not stated	–	55.6
V-Droid (Dai et al., 2025)	8B + LLM	>1	yes	AndroidWorld apps and tasks	verifier over candidates	59.5
GUI-Owl-7B (Ye et al., 2025)	7B	1	yes	not stated	–	66.4
Mobile-Agent-v3 (Ye et al., 2025)	32B	>1	yes	not stated	manager, worker, reflector, notetaker	73.3
MAI-UI-235B-A22B (Zhou et al., 2025)	235B	1	yes	not stated	none	76.7
MobileRL-9B (Xu et al., 2026)	9B	1	yes	online RL on generated task variants	–	80.2

Table 4. Step confidence $c = p(\hat{a} \mid s)$ on AndroidControl-High (8,444 steps). Risk@50: error rate on the 50% most confident steps. Last row: AUROC of the lowest step confidence in an episode for episode success.

	d1-2B (base)	DMA-2B (ours)
SR (%)	40.2	61.7
Mean confidence (%)	62.7	81.1
ECE ↓	0.226	0.194
AUROC ↑	0.721	0.758
Risk@50 (%) ↓	44.2	19.0
AURC ↓	0.420	0.196
Type: AUROC / ECE	0.668 / 0.059	0.800 / 0.073
Episode: AUROC	0.772	0.733

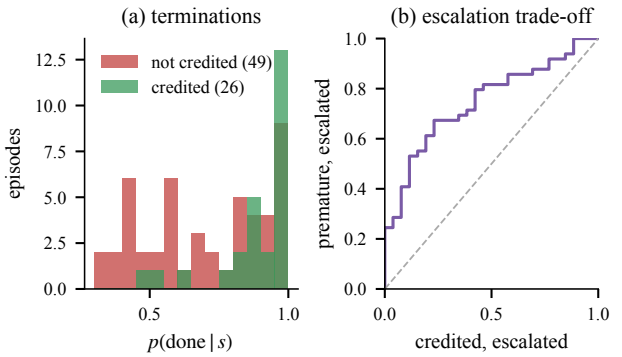


Figure 3. AndroidWorld terminations of DMA-2B. (a) Distribution of $p(\text{done} \mid s)$ for terminations that earn credit and for premature ones. (b) Share of each group that a gate escalates, as the threshold varies.

Control test set also contains no step whose correct action is "done", yet DMA-2B predicts "done" on 1,171 of the 8,444 steps, 460 of them on an episode's last step.

The 91 failed AndroidWorld tasks. Of the 91 tasks without full success, 51 ended with a premature termination (49 with no credit, 2 with half credit) and 40 exhausted the step budget. For each failure we located the first wrong step in the trace and the screenshots and checked the upstream success condition (Appendix A). About 36 failures, an estimate from the group-level reading, need abilities that a single pass under our constraints does not provide: arithmetic and date computation, memory across screens (counting, duplicates, transcription), text that exists only in pixels, colour or icon semantics without training labels, and state that the accessibility tree does not expose. These are the operations that TypeSafe assigns to code rather than to the model (TypeSafe AI, 2026b).

Harness and environment. With the checkpoint fixed, harness corrections raised the TSR from 15.5/115 to 25/115 (Appendix B): answers had not reached the checker, the accessibility tree lagged behind the screenshot, the SIM was unavailable, and first-run screens of Chrome, Markor and OsmAnd blocked tasks. Each defect produced failures that looked like model failures.

Vision. Replacing every screenshot with a uniform grey image lowers validation step accuracy from 519/827 to 337/827 and action-type accuracy from 80.0% to 50.8% (Appendix C); the model uses the image mainly to decide which kind of action to take.

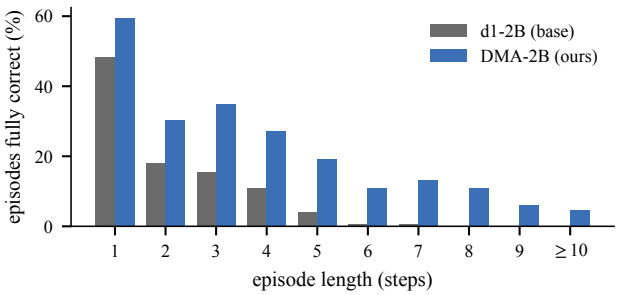


Figure 4. AndroidControl-High: share of episodes in which every step is correct, by episode length (1,543 episodes).

8. Limitations

All AndroidWorld numbers come from one seed and one attempt per task; a difference of five tasks is within noise (sign test $p \approx 0.2$ at five wins and one loss). The goal-to-app training rows use AndroidWorld's goal templates. The comparison rows in Tables 1 and 3 come from other papers and other harnesses. The conformal analysis assumes that calibration and deployment steps are exchangeable, which does not hold across scorers or splits (Section 6), and the termination thresholds are fitted in-sample. The failure taxonomy was produced by LLM agents reading traces, one per group, without a second annotator.

9. Conclusion

A 2B decision model that answers typed questions in one forward pass can operate a phone at 135 ms per decision and reaches 89.3% and 61.7% step success on AndroidControl-Low and High. Its action probability is over-confident but ranks its own errors well enough to act on about half of the steps at an error rate below 20%, and to flag most premature terminations on AndroidWorld. Task success (21.7%) is limited less by individual steps than by termination, by harness and environment defects, and by computation that a single pass should not be asked to perform. The next steps follow from these findings: verify progress on the next screen before terminating, re-rank candidate elements with per-candidate questions, and move counting, arithmetic and formatting into code that consumes the model's per-item judgments.

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A. Failure analysis of the AndroidWorld run

The 91 failed tasks were divided into four groups. One LLM agent per group, with no shared context, located the first wrong step from the trace and the screenshots and checked the upstream success condition. Counts are tasks.

Group (tasks)	First wrong step
Information retrieval (21)	answer visible but not reported (4); answer format (1); wrong navigation (4); date, time or colour reasoning (4); counting across screens (2); answer not expressible as a span (6)
Easy (27)	long press required (5); verification tasks already satisfied (3); premature termination (4); wrong element (4); target scrolled out of view (3); typing replaced instead of appended (2); clipboard chip absent from the element list (2); environment or app knowledge (4)
Medium and hard (34)	app procedure (7); values that must be computed, such as dates and end times (6); stopped after the first of several items (4); environment (3); form fields left empty (3); duplicates across screens (3); long note content (2); media library not loaded (2); single cases (4)
Visual (9)	navigation without reading (2); drawing or maze pages (2); text only in pixels (4); arithmetic (1)

B. Harness revisions

Same checkpoint and seed; paired comparisons are per task (sign test).

Revision	Score	Change with respect to the previous revision
A	15.5/115	–
B	20.5/115	answers forwarded to the checker; a fresh observation when the accessibility tree lags the screenshot (up to 8 s). 5 wins, 0 losses ($p = 0.062$).
C	25/115	fresh observation also after a page change (up to 20 s); working SIM; Chrome first-run screens cleared before the suite. 5 wins, 1 loss ($p = 0.22$).
D	pending	numeric answer format; Back closes the keyboard after typing; Markor and OsmAnd first-run state repaired. Only the affected tasks are re-run.

Environment defects found from traces: Markor opened on its first-run screens in all six Markor episodes on one emulator; a database query run as root on a missing OsmAnd database left an empty, root-owned file that OsmAnd could not open; VLC's media library remained at "Loading" on every emulator (cause inferred from the log); and running `uiautomator dump` stops AndroidWorld's accessibility forwarder.

C. Ablations and negative results

Validation split (827 steps)	Real screenshot	Grey screenshot
Step accuracy	519/827	337/827
Action-type accuracy	80.0%	50.8%
Element accuracy given the correct type	271/414	124/207
Type-text steps correct	37/53	3/53
Scroll steps correct	40/86	7/86

D. Artifacts

All numbers are recomputed from files in the project repository: AndroidControl per-step verdicts and confidences (`benchmarks/results/2026-11-12-calibration/conf-*-achigh.jsonl`), the calibration analysis (`benchmarks/train/calib_current.py`), AndroidWorld traces (`benchmarks/results/2026-11-12-maskenv/`), the difficulty and level tables (`benchmarks/aw/aw_report.py`), the figure and table scripts (`figures/make_figures.py`, `tables/make_ac_table.py`), and the AndroidWorld task metadata (`references/raw/androidworld-task_metadata.json`).

Attempt	Result
Threshold on $p(\text{done})$ that switches to the next action	harmful at every threshold on an earlier offline split
Separate "task done?" question	redundant: its label equals "action = done" on 137 of 137 validation rows
Masking an action already marked "no change"	2 wins, 2 losses on AndroidWorld; more premature terminations
Weight averaging (0.3/0.7) with an earlier checkpoint	6 wins, 9 losses on AndroidWorld
Recovery ("go back") rows	300-step pilot: validation step accuracy 65.2% $\rightarrow$ 57.7%; back predicted where it is wrong 1.9% $\rightarrow$ 10.0%
Per-candidate yes/no re-ranking of the top-5 elements, zero-shot	worse than the trained choice (261 vs. 286 of 430 validation targets), although the gold element is in the top 5 for 388
Next-screen progress question, zero-shot	AUROC 0.76 on an AndroidControl sample, but with only 26 negative transitions